

Varen 6869 AgSp1.1 cr	PGNNPGVEMGPDGQINKIILPSTPANPSSSGNPNQ..Q.....	36
Mg TRINITY DN1546 c0 g1 Cr	GGNTPGAELDPDGQIKKVIILPSTPSEPFSSSSLPSSQ.....DSQLNGREPYE	47
Msag 19329 AgSp1.1 cr	PGTTTPGVVTGSDG.....QPVK.....Y	18
Msag 18528 AgSp1.2 cr	PGTTTPGVVRIDPDGKINNVLPPISPNSELGGLNPDNEP.IPPFGPDNTPNSP.NFPGNYPRSTFKFPGFPGAPGSMGPIGFLDFSELS..ESPE	91
Varen 6869 AgSp1.2 cr	PGNTPGAELGPDGQINRIILPSTPPNPPSSGGLNPDGEP.SAPFGPGNSFNPSK.....SPGFKFPGFPGAPGSMGPIGFLDFSELS..ESPE	87
Mlaby 7048 AgSp1 cr	PGHTPGAELGPNNGNINRIFLPTAPNPPSPGGLNPPGEP.IAPFGPGNSPNSP.QSPGNYPGSAFQFPGFPGAPGSMGPIGFLDFSELS..ESPE	93
Aarg AgSp1 FC511 cr	PGTTPGAELGPDGKINKIIVLPTTPNPPPSGGLNPDNEP.IPPFGPGNSPNSP.QTPGNYPGSAFQFPGFPGAPGSMGPIGFLDFSELS..ESPE	93
Atri AgSp1 genomic cr	PGTTPGAELGPDGKINKIILPTTPNPPSSGGLNPKGEP.IPPFGPGNSPNSP.QPPGNYPGSAFQFPGFPGAPGANGPIGYYDFSQPLPSGESPE	93
Mhut 1271 AgSp1 cr	PGITPGVELGPDGHINRIVLPTPPPNPLFSDPLNPVKGK.AAPFGPGNIPNSP.YSPGNYPGSAFQFPGFPGAPGSMGPIGFLDFSELS..ESPE	93
Ncruc 58532and13532 AgSp1 cr	PGNTPGAEMGPDGHINRIVLPTSPNPQSSGPQYPGGGGP.GPFGPGNSPNSPG.QSPGNYPGSAFQFPGFPGAPGANGPIGYYDFSQPLPSGESPE	95
Amarm 50219 AgSp1.1 cr	PGNTPGAQLGPDGHINRIVLPTAPNPPSSGGLNPAEP.TAPFGPGNSPNSP.QSPGNYPGSAFQFPGFPGAPANGPIGYYDFSQPLPSGESPE	93
Lcorn 51676 AgSp1.1Cfrag cr	PGNTPGAELGPDGQITRIILPTPPNPPSSGGLNPAEP.IAPFGPGNSPNSP.QSPGNYPGSAFQFPGFPGAPGSMGPIGFLDFSQPLPSDVSP	93
Lcorn 58650 AgSp1.2 cr	PGNTPGAELGPDGQITRIILPTPPNPPSSGGLNPAEP.IAPFGPGNSPNSP.QSPGNYPGSAFQFPGFPGAPGSMGPIGFLDFSQPLPSDVSP	93

Varen 6869 AgSp1.1 cr	SFRPSGDPFSLNLGPDGSLNTDSIPDFDNPGYGSPPGYQVPLGYFDSVP.....NSDPSSQFVLPGSPGSPGSDGQFPFG	109
Mg TRINITY DN1546 c0 g1 Cr	MDDSVGHFPNFNLEIGGSFLGYEGGPNNLGLGENLNIETLPHYFIKPGSGFGSSHVPFHFHDYNQLPPDFNPNFPGLFALPGYSGFPESSRSQFAG	142
Msag 19329 AgSp1.1 cr	IAPPGGISPNFGLEIGGVFPSSISAGADDSAPGEHLNIHSTPNLINPLYGFPSSFAQAPLGIFYSSQ.....SSEFSDKFLVLPGDLLSPGSEGGQFLG	108
Msag 18528 AgSp1.2 cr	FDGSGIFFPNFNPEIGGFPFGFSGGPEDSGGGGLDIDSIPDFINPLYGFGSPQAPLGIFYDFNKLPPDFNSDFSGMFVLPGSPGSPGSSNGQFPFG	181
Varen 6869 AgSp1.2 cr	MYGSIQFLPNFSPEIGGFPFGFSGGPDGFGPGGYLNTDSLPDFINPGYGFPGSPQAPLGIFYDFNKLPPDFNSDFSGMFVLPGSPGSPGSSNGQFPFG	182
Mlaby 7048 AgSp1 cr	QVGSIGYFPDFSGEIGGFPFGFSGGPGGPGGGYLNIMSLPDFTNRGYGFPGSPQAPLGYLDESMLPKGFNPNFSGQFVYPGYPGSPGSGGGQYPG	188
Aarg AgSp1 FC511 cr	MEGSIGFLPNFSPEIGGFPFGFSGGPGGSPGLSGFLNNSLPDFINQGYGFPGSPQDPLGYLNFMSMLPPGYNPDFSQGFVYPGYPGSPGSGGGQFPFG	188
Atri AgSp1 genomic cr	MEGPPIGFLPNFSPEIGGFPFGFSGGPGGSSGGGGFLNNSLPDFINQGYGFPGSPQDPLGYLNFMSMLPPGYSPDFSQGFVYPGYPGSPGSGGGQFPFG	188
Mhut 1271 AgSp1 cr	MIGSIGFLPNFSPEIGGFPFGFSGGPDGSGGGYLNIDSLPDFNFKPGYGFPGSPQAPLGYLDKSKLPPGYNPFGSGQLVFPGYPGAPGSGSQFPFG	188
Ncruc 58532and13532 AgSp1 cr	MVGNIGFFPNFSPEIGGFPFGFSGGPDGSGGGFLNIGSLPNFKSPGYGFPGSPQAPLGYLQNMPLPPGYNPDFSGLPLVFPGYSGSPGSGGGQLPG	190
Amarm 50219 AgSp1.1 cr	MVGSIGFFPNFSPEIGGFPFGFSGGPGNSGGGGFLNINSLPDFMNPVYGFPGSPQAPLGYLDESMMLPPGFNLDSSGQLVFPDGPSPGSGSQFPFG	188
Lcorn 51676 AgSp1.1Cfrag cr	MVGPPIGFFPNFNPEIGGQFPFGFSGGPDGSGGGFLNINSLPDFRNPGYGFPGSPQAPLGYLDESMLPQDFDFDFSQGFVFPGYPGSPGSGSQFPFG	188
Lcorn 58650 AgSp1.2 cr	MVGPPIGFFPNFNPEIGGQFPFGFSGGPDGSGGGFLNINSLPDFMNPYGFPGSPQAPLGYLDESMLPQDFSLDFSQGFVFPGYPGSPGSGSQFPFG	188

Varen 6869 AgSp1.1 cr	GFLSFMQLPKSITDKINRTFDFTELIQAVQPLFPGNTINLGAIPRDNLQDIPGFDGSYNNLKFDYFGDSTKQNGGIFYIPELIIYLINQLPG....	200
Mg TRINITY DN1546 c0 g1 Cr	GFLSFPQLPKGITNNINGSFSFAELIQALQPLFPGKSNINVGAIQNKQLVMPGFNGTYDNLGLDIGDSSHNPNGGVFYLPVLVRLVGYLPVGSFG	237
Msag 19329 AgSp1.1 cr	GFLSFPQLSKDITNLINGSFSFAELIQALQPLFPGNSINVGAIHSELVMPGFNGTYDNLKLDNIKNNGHPNGGVFYLPVLVRLIDFVPVGSFN	203
Msag 18528 AgSp1.2 cr	GFLSFQQLPKSITDMINGTFSFQELIQALQPLFPGNSINVGAIQDKLQDIPGLKGNHDSKLKSLQLGDNSSQSGSLFYIPELMILISHLPD....	272
Varen 6869 AgSp1.2 cr	GFLSFPQLPKDITDKIDRPFSSFPQLIQAQPLFPGDSINVGAIPEDKLKDMPGFNGTSDSLRLDNFEDNNRPSGGVFYIPELVRLISYLPVGSFG	277
Mlaby 7048 AgSp1 cr	GFLSFSQLPESITNIHGSFSFSQMMNALGLFPGQ.VNIGAIPKNQLQSIPGFNGNYDNLMLDNYGDSQQPTGGVFYLPVLVRLASYPVGSFG	282
Aarg AgSp1 FC511 cr	GFMSFPELPKDVTDKINGTFSLPQLMQALQPLFPGQINIGAIPKQMQNIPGFEGTYDNLRLANFGNNNPVGGVFYLPVLVRLISYLPVGSFG	283
Atri AgSp1 genomic cr	GFLSFSLELPQGVTDKINGTFSLPQLMQALQPLFPGQINIGAIPKQMQNIPGFKGTYDNLRLANFGSDNQPTGGVFYLPVLVRLTSYLPVGSFG	283
Mhut 1271 AgSp1 cr	GFLSFQQLPKDITDKISGTFSFPQLINSLQPLFPGQINIGTIPQDKLQDIPGFTGTYNLKLANFGDSSQPTGGVFYLPVLVRLIGYLPVSSS	283
Ncruc 58532and13532 AgSp1 cr	GFLSFPQLPKDVTDKISSFTFSFPELKLQPLFPGQININMGAIPQDKLQDIPGFNGTYDNLKLANYGDNQPTGGVFYLPVLVRLMSYLPVGSFG	285
Amarm 50219 AgSp1.1 cr	GFLSFPQLPKDITDKIDGTFSFPQLIEALQPLFPGQINIGAIPKDKLQDIPGFEGTYDNLKLANFGGNNQPTGGVFYLPVEMVRLIGYLPVGSFG	283
Lcorn 51676 AgSp1.1Cfrag cr	GFLSFPQLPKDITDKINGTFSFPQLINALQPLFPGQINIGAIPQDKLQDIPGFNGSYDNLKLANFGDSSQPTGGVFYLPVLVRLISYLPVGSFG	283
Lcorn 58650 AgSp1.2 cr	GFLSFPQLPKDITDKISGTFSFPQLINALQPLFPGQINIGAIPQDKLQDIPGFNGSYDNLKLANFGDSSQPTGGVFYLPVLVRLISYLPVGSFG	283

Varen 6869 AgSp1.1 cr	GISGT F DFPGFYGPRRN I CDAG N ...D S SDSGSQDLGGEVE G DL D GLFGDMGDAAPSDTMD A APR.....	263
Mg TRINITY DN1546 c0 g1 Cr	IGPGTINQ E GGFGDAIK F PGLNGAPGYICDYP E NG...DATDRGSQDLGEGVFGDGNTD T ND D CFQES.....	302
Msag 19329 AgSp1.1 cr	K GLGTINQ E GGFGHPV N FPGLS G TLGYICD Y L E NG...DATIEGSQIFS.....	249
Msag 18528 AgSp1.2 cr	IGPETINSDGGF S TPFN F PGFS G PPGYICDYPDKD...DVT D GGFHD L GGG V KGN D NR P FGDV G GAAPDEDD A NDVAS Q DLGGEV K .INKTGPF R D	364
Varen 6869 AgSp1.2 cr	K GS G TINQDGGFG E PFNF P GLNGAPGN I CDY P ED Y YD Y SMDEGSQDLGGEVE G DNG P SGDV G DAAPED I MGAAP P NNL S P.....	358
Mlaby 7048 AgSp1 cr	 A NGGFGDP F GE Q LL H GAPGYICDYPDND...DTTAGGSQDLGGE I KGN G HGPSGDVADAAP... N LGSAPGND L GGAAP GNT	359
Aarg AgSp1 FC511 cr	NGPGTINQDGGFGDPFN F PGLNGAPGYICDYPDNG...DAATGRSQ N LGGE I KGN G NGPSGDVADAAP E AD L GGAAP Q SD L A.....	362
Atri AgSp1 genomic cr	NGPGTINQNGGFGDPFN F PGLNGAPGYICDYPDNG...HATTGGSQ N LGGE I KGN D NGPSGDVADAAPGN N LGAAPGN N LGAAP GN.	368
Mhut 1271 AgSp1 cr	K DSE I INK N GGF S N PFNF P GLNG F PGYICDYPDNG...DATTEGSQDLGGE I EG Y GNG P FEDVADAAP E NDLE.....	353
Ncruc 58532and13532 AgSp1 cr	K GPGTINHDGGF D DPFN F PGLNGAP S ICDYPDNG...DATDGGSQDLGGEV K DGN P AGDVADAAP... D LGA P APGND L GGAAP D LGA P APGND	376
Amarm 50219 AgSp1.1 cr	YGPGTINHDGGFG H PFNF P GLNGAPGYICDYP N NG...VL...SQDLGGEVKGN G NGPSGDVADAAP E ND L G G TAP E ND I GGAAP EN.	364
Lcorn 51676 AgSp1.1Cfrag cr	K GS G TINQDGGFGDPFN F PGLNGAPGYICDYPDNG...DATAGGSQDLGGEVKGN G NGPSGDVADAAP... D LGS P AP E ND L GGAAP.....	364
Lcorn 58650 AgSp1.2 cr	K GS G TINQDGGFGDPFN F PGLNGAPGYICDYPDNG...DATAGGSQDLGGEVKGN G NGPSGDVADAAP... D LGS P AP E ND L GGAAP.....	364

Varen 6869 AgSp1.1 cr	GKEVKDNRNGSSGDVGNSFPDYS I V G VTPL T N Q S.....SG A	263
Mg TRINITY DN1546 c0 g1 Cr	GKEVKDNRNGSSGDVGNSFPDYS I V G VTPL T N Q S.....SG A	339
Msag 19329 AgSp1.1 cr	GKEVKDNRNGSSGDVGNSFPDYS I V G VTPL T N Q S.....SG A	249
Msag 18528 AgSp1.2 cr	VGGAAPNEDGANDGGSQDLGGEV.KSNKVG P FRD V G G AAPYEDGANDGDSQDLGGEVKGSGNGPSGDVADAAPDDN M ADAP Q S N L..SS P T P V T	456
Varen 6869 AgSp1.2 cr	GKEVKDNRNGSSGDVGNSFPDYS I V G VTPL T N Q S.....SG A	359
Mlaby 7048 AgSp1 cr	 L GGS A PG..N..... T LGS S AP.....GNAP G GSAPGN N M..A S N T L V G	394
Aarg AgSp1 FC511 cr	GKEVKDNRNGSSGDVGNSFPDYS I V G VTPL T N Q S.....SG A	368
Atri AgSp1 genomic cr	 N LGS A A..... P GN N LGS A AP.....END L GGAAP E NN P AP Q S D T P SS	406
Mhut 1271 AgSp1 cr	GKEVKDNRNGSSGDVGNSFPDYS I V G VTPL T N Q S.....SG A	368
Ncruc 58532and13532 AgSp1 cr	LGGAAPD L GAP A PG..NDLGAAPDLGAP A PGND L GGAAPDLGAP A P..GNDLGGAAADLG.....APAPAGND L GATDPP S D Q ..G S N T L V A	459
Amarm 50219 AgSp1.1 cr	DLGG A A..... P END L GGAAP.....END L GGAAP Q SEMG..S N T L V A	400
Lcorn 51676 AgSp1.1Cfrag cr	DLGSP A P END L GGV A P.....NLGSPAP Q T N Q G ..S N T L V A	398
Lcorn 58650 AgSp1.2 cr	DLGSP A P END L GGV A P.....NLGSP A SE T N Q G..S N T L V A	398

Varen 6869 AgSp1.1 cr	.NDPSCGDDVNS A F S T A K Q AL Q	284
Mg TRINITY DN1546 c0 g1 Cr	DIDG N CD N DVATT F NT T K Q A I R	361
Msag 19329 AgSp1.1 cr	NDPSCGDDVNS A F S T A K Q AL Q	249
Msag 18528 AgSp1.2 cr	GT E G N CEDDVDRTLNT A K Q AL Q	478
Varen 6869 AgSp1.2 cr	TSVTL C EDNVVST F S V A K QAL Q	381
Mlaby 7048 AgSp1 cr	P EDEGCQDNVPEAF S KARQAL A	416
Aarg AgSp1 FC511 cr	PTD Q DCEEDVHGAF S KARQAL M	390
Atri AgSp1 genomic cr	PTDK G CEEDVQGA F S K ARQAL M	428
Mhut 1271 AgSp1 cr	PEVKDCQDSVPEAF R KAR H AL T	390
Ncruc 58532and13532 AgSp1 cr	PDEEDCQDNVPEAF H KARQAL A	481
Amarm 50219 AgSp1.1 cr	PEDEDCQDNVPEAF N KARQA T	422
Lcorn 51676 AgSp1.1Cfrag cr	PDEEDCQDNVPET F S K AREAL A	420
Lcorn 58650 AgSp1.2 cr	P D N EDCQDNVPET F S K AREAL A	420

X non-conserved
X similar
X ≥ 50% conserved